

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems. (BL6)*

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

Dr.

QUESTIONS: 1 (1 X 100 = 100)

Total Marks: 100 Duration: 3hrs

1. Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications. (BL6)
2. Develop a cloud-edge communication model for wireless healthcare monitoring. (BL6)
3. Design a functional architecture for real-time edge data processing in industrial automation. (BL6)
4. Propose a cloud-based wireless framework for smart surveillance systems. (BL6)
5. Design an edge AI workflow for intelligent wireless data filtering. (BL6)
6. Develop a task offloading strategy between edge and cloud in 5G applications. (BL6)
7. Design a low-latency communication architecture for edge-assisted smart city services. (BL6)
8. Propose a scalable edge-cloud framework for wireless sensor networks. (BL6)
9. Design a Cloud Radio Access Network model for dense urban deployment. (BL6)
10. Develop a functional split model between centralized and distributed radio units. (BL6)
11. Design a virtualized wireless access framework for next-generation communication. (BL6)
12. Propose a data flow model for cloud-based radio resource allocation. (BL6)
13. Design a centralized traffic control model using Cloud RAN principles. (BL6)
14. Develop a network management architecture for cloud-assisted wireless systems. (BL6)
15. Design a wireless service framework integrating virtual network functions. (BL6)
16. Propose an optimized fronthaul communication model. (BL6)
17. Design a satellite-assisted communication model using Low Earth Orbit satellite for remote wireless coverage. (BL6)
18. Develop a comparative architecture for LEO, MEO, and GEO backhaul systems. (BL6)

19. Design a spectrum sharing model between LTE and 5G users. (BL6)
20. Propose a frequency allocation strategy for mixed wireless services. (BL6)
21. Design a dynamic spectrum access framework for urban wireless systems. (BL6)
22. Develop a communication architecture combining terrestrial and satellite networks. (BL6)
23. Design a spectrum efficiency improvement model for dense communication zones. (BL6)
24. Propose a channel reuse plan for multi-user wireless communication. (BL6)
25. Design an NB-IoT architecture for smart utility monitoring. (BL6)
26. Develop an LTE-M communication workflow for wearable monitoring systems. (BL6)
27. Design a low-power communication model for battery-operated IoT devices. (BL6)
28. Propose a device communication plan for long-life sensor networks. (BL6)
29. Design a 5G-enabled IoT framework for campus automation. (BL6)
30. Develop an energy-efficient wireless communication strategy for sensor nodes. (BL6)
31. Design an IoT communication flow for periodic environmental sensing. (BL6)
32. Propose a scalable wireless architecture for smart connected devices. (BL6)
33. Design a smart city communication framework for traffic monitoring and surveillance. (BL6)
34. Develop an energy monitoring communication model for public infrastructure. (BL6)
35. Design a smart factory information flow for predictive maintenance. (BL6)
36. Propose a robotic automation communication framework for industrial control. (BL6)
37. Design a remote patient monitoring communication architecture. (BL6)
38. Develop a wireless communication workflow for AI-assisted healthcare diagnosis. (BL6)
39. Design a smart irrigation communication model for crop monitoring. (BL6)
40. Propose a livestock tracking communication architecture using wireless sensors. (BL6)

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Design Task					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation

1. Task:

To design and simulate a wireless service framework that integrates Virtual Network Functions (VNFs) for dynamically managing wireless network services such as routing, security, load balancing, and Quality of Service (QoS).

2. Description:

- Traditional wireless networks use dedicated hardware for functions such as routing, firewalling, traffic management, and load balancing. This makes the network expensive and difficult to modify.
- Network Function Virtualization (NFV) solves this problem by implementing these network functions as software running on general-purpose computing platforms.
- The proposed framework consists of:
Wireless Users → Base Station → VNF Manager → Virtual Network Functions → Internet/Cloud
- The main VNFs used in the proposed design are:
 - VNF-1: Firewall – filters unwanted traffic.
 - VNF-2: Load Balancer – distributes traffic among available servers.
 - VNF-3: QoS Manager – prioritizes traffic according to service requirements.
 - VNF-4: Router – forwards packets toward the destination.
- The VNF Manager monitors network traffic and dynamically selects the appropriate virtual functions.

3. Design Parameters:

Parameter	Symbol	Description	Typical Design Range
Active user capacity per edge node	N_{user}	Number of active users (UEs) that can be supported by one edge node.	$10^3 - 10^5$ UEs
Compute capacity required per VNF instance	C_{vCPU}	Virtual CPU resources required by one instance of a VNF for processing the traffic.	2 – 64 vCPUs
Fronthaul / Backhaul bandwidth requirement	B_{link}	Bandwidth required between base station and edge/cloud (VNF hosting locations).	10 – 100 Gbps
Maximum allowable E2E delay budget	L_{max}	Maximum end-to-end latency allowed for the service to meet QoS requirements.	1 ms – 10 ms
Service processing rate of VNF node	μ_{service}	Packet processing rate (service rate) of a VNF instance or node.	1000 – 50000 packets/sec
Incoming packet arrival rate	λ_{traffic}	Average packet arrival rate at the VNF (assumed Poisson process).	Poisson process (pks/sec)

4. Mathematical Formulations & Design Equations:

A. End-to-End Latency Budget Calculation:

The total end-to-end delay T_{E2E} across a Service Function Chain (SFC) consisting of K chained VNFs is given by:

$$T_{\text{E2E}} = T_{\text{wireless}} + \sum_{i=1}^K (T_{\text{proc},i} + T_{\text{queue},i}) + \sum_{j=1}^{K-1} T_{\text{prop},j}$$

Where:

- T_{wireless} = Over-the-air transmission latency (symbol duration + TTI).

- $T_{\text{proc},i} = \frac{D_i}{f_i}$ = Processing delay of i^{th} VNF (where D_i is task complexity in cycles and f_i is allocated vCPU frequency).
- $T_{\text{queue},i} = \frac{1}{\mu_i - \lambda_i}$ = Queuing delay using an $M/M/1$ queue model ($\mu_i > \lambda_i$).
- $T_{\text{prop},j} = \frac{d_{j,j+1}}{v}$ = Propagation delay over distance d through fiber/transport network ($v \approx 2 \times 10^8$ m/s).

B. VNF Placement and Resource Optimization Problem:

To minimize total power and cloud infrastructure deployment cost subject to latency and capacity constraints:

$$\min_{x_{i,n}} \sum_{n \in \mathcal{N}} P_n \cdot y_n + \sum_{i \in \mathcal{V}} \sum_{n \in \mathcal{N}} c_{i,n} x_{i,n}$$

Subject to Constraints:

1. **Placement Constraint:**

$$\sum_{n \in \mathcal{N}} x_{i,n} = 1, \quad \forall i \in \mathcal{V}$$

2. **Capacity Constraint:**

$$\sum_{i \in \mathcal{V}} x_{i,n} \cdot R_i \leq C_n, \quad \forall n \in \mathcal{N}$$

3. **Latency Constraint:**

$$T_{\text{E2E}} \leq T_{\text{max}}$$

Where $x_{i,n} \in \{0,1\}$ indicates if VNF i is hosted on physical server n ; R_i is resource demand; C_n is node capacity; P_n is idle power.

5.Design Procedure:

➤ **Define the user capacity**

Select the required number of active users based on the target service area. For example, choose UEs per edge node.

➤ **Estimate traffic arrival rate**

The incoming traffic is modeled as a Poisson process:
where is the average packet rate per user.

➤ **Determine VNF processing capacity**

Select the VNF service rate such that:

This prevents continuous queue growth and ensures stable VNF operation.

➤ **Calculate the required number of VNF instances**

where is the desired maximum utilization, typically around 0.7–0.8.

➤ **Allocate computing resources**

Each VNF instance is assigned virtual CPUs according to its processing requirement. The total computing requirement is:

➤ **Design the fronthaul/backhaul link**

Select sufficiently higher than the expected traffic load so that the link does not become a bottleneck.

➤ **Check the latency requirement**

The total service-chain delay is:
and the design must satisfy:

➤ **Finalize the VNF service chain**

Based on these calculations, the VNFs are arranged as:

Wireless Users → Edge Node → Firewall VNF → QoS VNF →
Load Balancer VNF → Router VNF → Cloud/Internet

➤ **Performance verification**

Finally, evaluate VNF CPU utilization, link utilization, packet processing rate, end-to-end delay, packet loss and throughput using MATLAB.

6. MATLAB Implementation:

```
clc;
clear;
close all;

N = 1000;

packetSize = randi([500 1500],1,N);
traffic = packetSize .* 8;

priority = randi([1 3],1,N);
malicious = rand(1,N) < 0.08;

firewallPassed = ~malicious;

qosWeight = zeros(1,N);

qosWeight(priority == 1) = 1.0;
qosWeight(priority == 2) = 0.8;
qosWeight(priority == 3) = 0.5;

serverLoad = [0 0];

processedPackets = 0;
totalDelay = 0;

serverAssignment = zeros(1,N);

for i = 1:N

    if firewallPassed(i)

        if serverLoad(1) <= serverLoad(2)
```

```

        server = 1;
    else
        server = 2;
    end

    serverAssignment(i) = server;

    serverLoad(server) = serverLoad(server) + traffic(i);

    processingDelay = 2 + 5*qosWeight(i);
    routingDelay = 3;

    totalDelay = totalDelay + processingDelay + routingDelay;

    processedPackets = processedPackets + 1;
end
end

simulationTime = 1;

throughput = sum(traffic(firewallPassed)) / simulationTime / 1e6;

packetLoss = (N - processedPackets) / N * 100;

averageDelay = totalDelay / processedPackets;

utilization1 = serverLoad(1) / max(serverLoad) * 100;
utilization2 = serverLoad(2) / max(serverLoad) * 100;

fprintf('Wireless NFV Framework Results\n');
fprintf('-----\n');
fprintf('Total Packets      : %d\n',N);
fprintf('Packets Processed    : %d\n',processedPackets);
fprintf('Packets Blocked      : %d\n',N-processedPackets);
fprintf('Throughput           : %.2f Mbps\n',throughput);
fprintf('Packet Loss          : %.2f %%\n',packetLoss);
fprintf('Average Delay        : %.2f ms\n',averageDelay);

```

```
fprintf('Server 1 Utilization  : %.2f %%\n',utilization1);  
fprintf('Server 2 Utilization  : %.2f %%\n',utilization2);
```

```
figure;  
bar([utilization1 utilization2]);  
grid on;  
xlabel('Virtual Server');  
ylabel('Utilization (%)');  
title('Load Balancer VNF Utilization');  
xticklabels({'Server 1','Server 2'});
```

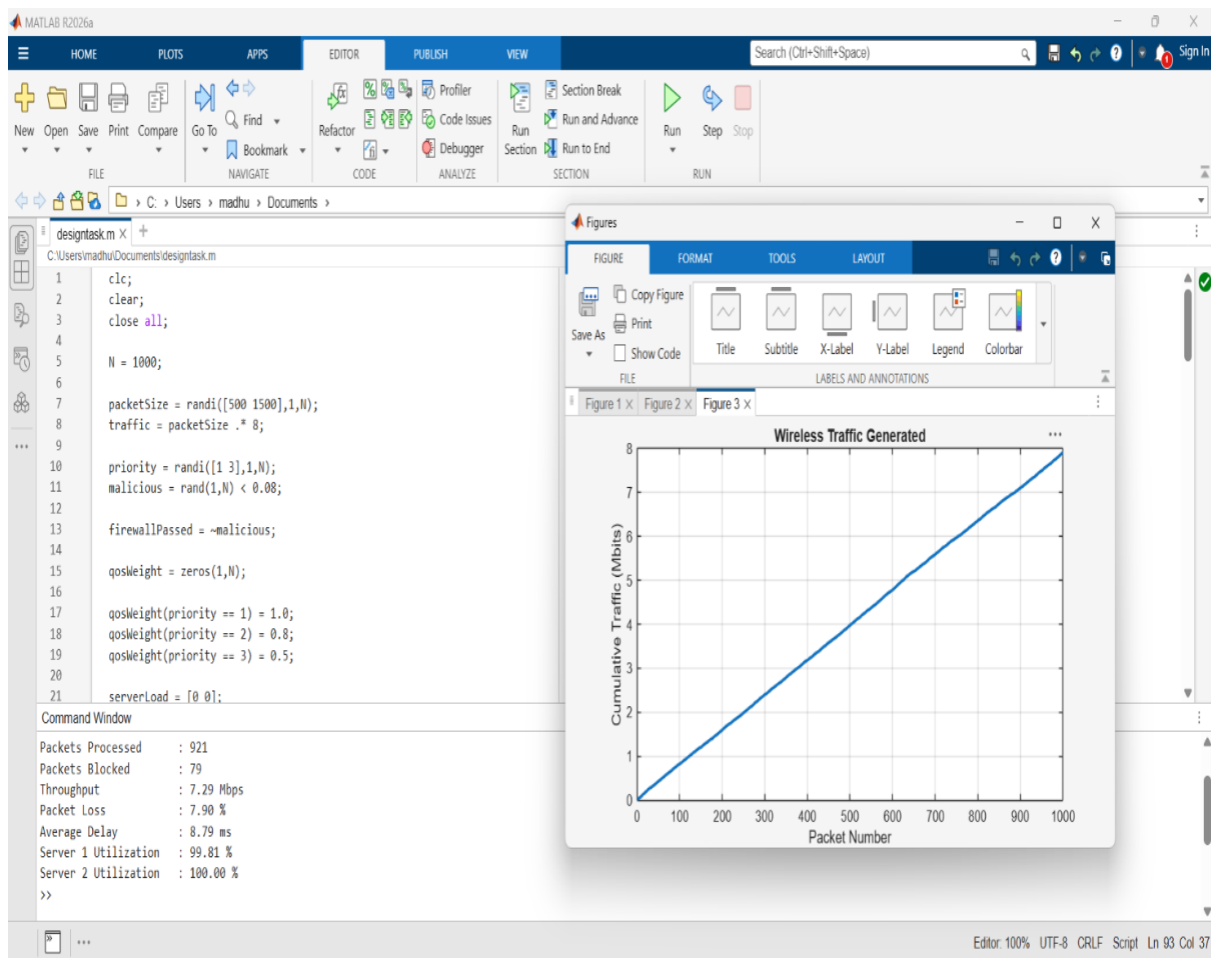
```
figure;  
histogram(priority);  
grid on;  
xlabel('Traffic Priority');  
ylabel('Number of Packets');  
title('QoS Traffic Classification');
```

```
figure;  
plot(cumsum(traffic)/1e6,'LineWidth',2);  
grid on;  
xlabel('Packet Number');  
ylabel('Cumulative Traffic (Mbits)');  
title('Wireless Traffic Generated');
```

7. Design Outcome:

- The proposed framework demonstrates how NFV can make a wireless network flexible and software-defined.
- Instead of requiring separate physical hardware for firewall, routing, QoS, and load balancing, these functions are implemented as software-based VNFs.

- The framework can therefore dynamically manage wireless traffic and improve resource utilization.



8.Summary:

- The wireless service framework is designed using user capacity, traffic rate, VNF processing capacity, vCPU resources, link bandwidth, and delay constraints.
- Suitable VNF instances are allocated based on traffic demand, forming a service chain of Firewall, QoS, Load Balancer, and Router VNFs.
- MATLAB is used to evaluate utilization, delay, throughput, packet loss, and scaling performance.

